

How to compute the Jaccard Similarity in this example? (Jaccard vs. Cosine)

Asked 5 years, 10 months ago Modified 5 years, 10 months ago Viewed 24k times



I am trying to understand the difference between Jaccard and Cosine. However, there seem to be a disagreement in the answers provided in [Applications and differences for Jaccard similarity and Cosine Similarity](#).



I am seeking if anyone could step me through the calculations of the Jaccard Similarity in this Cosine Similarity example from <https://bioinformatics.oxfordjournals.org/content/suppl/2009/10/24/btp613.DC1/bioinf-2008-1835-File004.pdf>

Given:

Supplement 1: An example for computing cosine similarity of annotations

$$\cos(\vec{t}_1, \vec{t}_2) = \frac{\vec{t}_1 \cdot \vec{t}_2}{\|\vec{t}_1\| \|\vec{t}_2\|} \quad (1)$$

To calculate cosine similarity between two texts t_1 and t_2 , they are transformed in vectors as shown in the Table 1. Each word in texts defines a dimension in Euclidean space and the frequency of each word corresponds to the value in the dimension. Then, the cosine similarity is measured by using the word vectors as in equation 1. For example, a cosine similarity can be computed as below for two texts: “*Glutathione homocystine transhydrogenase*” and “*Glutathione CoA glutathione transhydrogenase*.”

$$\frac{1 \cdot 2 + 1 \cdot 0 + 0 \cdot 1 + 1 \cdot 1}{\sqrt{1^2 + 1^2 + 0^2 + 1^2} \sqrt{2^2 + 0^2 + 1^2 + 1^2}} \simeq 0.72$$

	glutathione	homocystine	coa	transhydrogenase
$\vec{t}_1$	1	1	0	1
$\vec{t}_2$	2	0	1	1

Table 1: Each unique word in texts become a coordinate in Euclidean space. From texts t_1 and t_2 , “glutathione”, “transhydrogenase” in both t_1 and t_2 , “homocystine” in t_1 and “coa”, “glutathione” from t_2 are used as dimensions. The frequency of each word become a value for each dimension in a vector $\vec{t}_1$ and $\vec{t}_2$.

Question: **How do we compute the Jaccard Similarity index between t_1 and t_2 ?**

Thank you.

similarity

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edited Apr 13, 2017 at 12:50



Community

Bot

1

asked Dec 21, 2016 at 23:50



jkyh

462

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Sorted by:

Highest score (default)



1 Answer



5



Cosine similarity is for comparing two *real-valued vectors*, but Jaccard similarity is for comparing two *binary vectors (sets)*. So you cannot compute the standard Jaccard similarity index between your two vectors, but there is a generalized version of the Jaccard index for real valued vectors which you can use in this case:

$$J_g(a, b) = \frac{\sum_i \min(a_i, b_i)}{\sum_i \max(a_i, b_i)}$$

So for your examples of $t_1 = (1, 1, 0, 1)$, $t_2 = (2, 0, 1, 1)$, the generalized Jaccard similarity index can be computed as follows:

$$J(t_1, t_2) = \frac{1+0+0+1}{2+1+1+1} = 0.4$$

Alternatively you can treat your bag-of-words vector as a binary vector, where a value 1 indicates a words presence and 0 indicates a words absence *i.e.*

$t_1 = (1, 1, 0, 1)$, $t_2 = (1, 0, 1, 1)$. From there, you can compute the original Jaccard similarity index:

$$J(t_1, t_2) = \frac{2}{2+1+1} = 0.5$$

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edited Dec 22, 2016 at 1:58

answered Dec 22, 2016 at 0:56



timleathart

3,810

19

35

1 @jkyh please see my edit to this answer -- I misinterpreted your situation initially and have added extra clarification – [timleathart](#) Dec 22, 2016 at 2:02